VENKATESH MUDALIAR

Applied Data Scientist | Machine Learning Engineer

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SUMMARY

Applied Data Scientist / Machine Learning Engineer with 4+ years taking ML from problem definition through production deployment, measurement, and iteration. M.S. Data Science (Stevens Institute of Technology, GPA 3.82). Hands-on across the full ML lifecycle — feature pipelines, forecasting and recommendation models, A/B testing, and MLOps (Airflow, MLflow, Databricks) — deployed on AWS/GCP, with applied GenAI/LLM and agentic-workflow experience translated into measurable, shipped product impact.

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, Bash, REST APIs, Git

ML & Modeling: Scikit-learn, XGBoost, PyTorch, TensorFlow, Forecasting, Recommendation Systems, Ranking, Optimization, Regression, Classification, Clustering, Ensemble Methods, A/B Testing, Time Series

GenAI / LLMs: LangChain, RAG Pipelines, FAISS, Vector Databases, Embeddings, LLM-as-a-Judge, Agentic Workflows, Prompt Engineering

MLOps & Cloud: AWS (S3, EC2, SageMaker, Bedrock), GCP (Dataproc, GCS), Databricks, Docker, MLflow, Airflow, Snowflake, FastAPI

Visualization: Power BI, Tableau, Plotly Dash, Matplotlib, Seaborn

PROFESSIONAL EXPERIENCE

AI/ML Engineer (Research) | Stevens Institute of Technology *May 2025 – Sep 2025*

- Designed and shipped end-to-end ML evaluation pipelines benchmarking regression, classification, and LLM-based forecasting models on financial datasets; cross-validated feature selection improved F1-score ~12% and cut experimentation overhead ~30%.
- Built FAISS-based semantic search and embedding pipelines for multi-source retrieval, improving information-retrieval precision ~18% across multi-experiment benchmarks — directly informing model-selection trade-offs.
- Evaluated bias-variance trade-offs across traditional ML vs. LLM systems, producing structured benchmarks that guided downstream production model selection for forecasting use cases.

Data Scientist | Accenture *Feb 2023 – Sep 2024*

- Engineered Python/SQL feature pipelines for predictive modeling on large-scale financial datasets, reducing data preparation time ~25% and accelerating ML deployment cycles across 3 business units.
- Built automated anomaly detection and statistical validation frameworks that improved data quality 40%, increasing ML training-data reliability and model performance consistency.
- Delivered scalable data and feature-pipeline workflows supporting 5+ enterprise ML/analytics initiatives end-to-end — from pipeline design through stakeholder adoption.

Data Engineer | LTIMindtree Ltd. *Sep 2020 – Feb 2023*

- Engineered ETL and data-integration pipelines (Python, SQL, REST APIs) across SAP S/4HANA, automating workflows processing millions of records monthly and cutting manual effort ~30%.
- Built automated monitoring and real-time validation frameworks that reduced recurring data-quality issues 45%, enabling proactive resolution before downstream model or product impact.

PROJECTS

Financial Portfolio Prediction — Forecasting & Trading Signals | *github.com/Venkyyy98/Financial-Portfolio-Prediction*

- Combined FinBERT sentiment analysis with PySpark MLlib models on GCP Dataproc to generate Buy/Hold/Sell trading signals; achieved F1-score of 0.82 and validated performance via end-to-end backtesting against a passive benchmark — a forecasting/decision-intelligence pipeline directly analogous to production recommendation systems.

RagProbe — LLM Evaluation & RAG Security Benchmarking | *github.com/Venkyyy98/Ragprobe*

- Built an adversarial RAG framework (FAISS, LangChain, LLM-as-a-Judge) with automated evaluation pipelines, reducing hallucination 22% and improving context recall 18% across 500+ test cases — applied GenAI/agentic-workflow experience.

SAP CPI Reliability Studio — AI-Powered Operations Monitoring | *github.com/Venkyyy98/cpi-log-analytics-dashboard*

- Built an end-to-end observability platform (Streamlit, FastAPI) with anomaly detection and forecasting models to automate root-cause analysis and predict integration failures.

CERTIFICATIONS & EDUCATION

Certifications: AWS Certified AI Practitioner (2025–2028) | IBM Data Science Professional Certificate (2025) | Databricks Generative AI Fundamentals (2025–2027)

Education: M.S. Data Science, Stevens Institute (GPA 3.82, 2026) | B.E. Electronics & Telecom, Mumbai (2020)